

REVIEW

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Machine learning approaches for wind power forecasting: a comprehensive review

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Abstract

The dependable integration of wind energy into contemporary electrical systems, which supports market operations, grid stability, and strategic planning, depends on accurate wind power forecasting. A thorough analysis of machine learning (ML) techniques for wind power prediction is presented in this research, encompassing advancements from 2006 to 2025. Physical, statistical, traditional machine learning, deep learning, ensemble, and hybrid models are the categories into which current forecasting techniques fall. With particular focus on sophisticated designs like Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), and hybrid CNN–LSTM frameworks, we examine their capabilities, constraints, and application domains. The paper also looks at lightweight solutions that allow deployment on low-cost edge devices, as well as optimization algorithms like the Fruit Fly Algorithm and Particle Swarm Optimization (PSO). Comparative analyses show how well various models handle temporal dependencies, nonlinearity, scalability, and interpretability over a range of predicting horizons. There is a thorough discussion of the main obstacles, including problems with data quality, computing limitations, and performance in harsh weather. The findings indicate that in order to improve prediction accuracy, robustness, and real-time application, future research should concentrate on IoT-enabled sensor networks, multi-model fusion, physics-informed learning, and sophisticated structures like Transformers. All things considered, this paper charts the developing field of machine learning-driven wind power forecasting and offers practical guidance for developing intelligent, efficient, and sustainable renewable energy systems.

Keywords CNN, LSTM, Machine learning, Deep learning, Hybrid models, Optimization methods, Edge computing, Wind power forecasts

1 Introduction

Wind power facilities produce better environmental outcomes than fossil fuel-powered systems, and they are crucial parts of sustainable clean energy generation. Installations of wind power both benefit the environment and lower greenhouse gas emissions [1]. Strict energy policy regulations and the development of modern wind power technology are two ways that several national forces are combating climate change and promoting the rapid growth of the wind industry. Over the past 20 years, wind power plants have



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rapidly expanded globally, advancing various energy choices and reducing CO₂ levels [2]. Wind energy's intrinsic instability and sporadic nature provide major scheduling and grid control issues, necessitating the use of highly developed, precise wind power forecasting services [3]. For safe and reliable operation, power systems require precise wind power predictions. Wind power's energetic uncertainty raises the pressure on grid management systems for load distribution and scheduling [4]. Three main issues are caused by poor forecast data: unstable frequency regulation, a greater demand for expensive and polluting backup power systems, and catastrophic damage to power networks [5]. Various wind power forecast times correspond with operational requirements, such as power market trading, operational scheduling, and real-time control implementation. The availability of complete datasets and significant prediction accuracy is hampered by the numerous uncontrollable affecting factors, such as changing wind speeds and weather patterns [6]. Better processing power and more accessible datasets led to the widespread adoption of modern prediction techniques based on cutting-edge machine learning technologies [7]. Since machine learning can identify complex nonlinear patterns in addition to erratic wind power data, it performs better than statistical and physical models. By identifying lengthy pattern combinations and minute variations in wind speed and power data, the Long Short-Term Memory (LSTM) networks leverage deep learning technology to generate the most accurate time-series forecasts [8]. Because they provide excellent generalization qualities and consistently high predictive precision, the ensemble learning techniques Random Forest and XGBoost (Extreme Gradient Boosting) are often used as wind power prediction solutions. To improve wind power forecasting solutions that increase power grid dependability and reduce costs, scientists use many machine learning models in a single system [9]. In-depth discussion and evaluation of artificial intelligence algorithms for wind power forecasting are provided in this document. Through application evaluation, limitation analysis, and possible future research prospects, this work conducts a thorough examination of machine learning models.

1.1 Explicit distinction and role of machine learning (ML) vs. deep learning (DL) in wind power forecasting

1.1.1 Theoretical foundations

Machine Learning (ML) is often defined in a broad sense as a pattern learning approach to make predictions or decisions, without being explicitly programmed. The linear regression, decision trees, random forests, support vector machines (SVMs), and artificial neural networks (ANNs) belong to the ML techniques. These tend to be engineered algorithms and are useful in finding patterns when you have moderate-sized structured data. Specialized subsets of ML Deep Learning. Deep Learning (DL), its own subset that applies the use of multi-layered neural network architectures that have faculties, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory Networks (LSTMs), and Gated Recurrent Units (GRUs). DL models are able to automatically learn the hierarchical features of raw and high-dimensional data. They have a unique structure that allows powerful representation learning, especially when dealing with large and complicated datasets, as is the case with wind power forecasting.

Although classical ML algorithms are categorized under the greater umbrella of deep learning, the ability of DL to automatically implement feature extraction and correlate nonlinear complex relationships between them is what makes DL distinct relative to the traditional ML algorithms.

1.1.2 Practical implications in wind power forecasting

The choice between ML and DL approaches in wind power forecasting is shaped by several key factors:

- **Data Dependency**

ML algorithms (e.g., Decision Trees, Random Forests, SVMs) are well-suited to smaller or moderately sized datasets, often requiring manual feature extraction and selection. Architectures (e.g., LSTM, CNN) thrive on large-scale, high-resolution datasets such as those from SCADA systems. Their performance and generalization improve significantly with increased data volume and diversity.

- **Computational Demands:**

ML methods generally require less computational power and can be deployed on standard computing platforms, making them feasible for applications in resource-limited settings or for real-time edge deployment. DL models, due to their deeper and more complex architectures, are computationally intensive—requiring specialized hardware such as GPUs for training and inference. This is particularly evident with models like CNNs, LSTMs, and hybrid CNN-LSTM networks.

- **Model Interpretability:**

Classical ML models, especially tree-based methods (e.g., Decision Tree, Random Forest), and linear models (Linear Regression), offer high interpretability. This transparency aids operational decision-making and fosters trust among stakeholders.

DL models tend to operate as ‘black boxes’ with limited interpretability. Although efforts such as attention mechanisms and visualization techniques exist to improve explainability, model complexity remains a challenge.

1.1.3 Real-world deployment and operational considerations

Deployment of DL models in operational wind farms, especially on low-cost devices (e.g., Raspberry Pi), faces notable challenges: limited memory, processing power, and energy efficiency favor lightweight ML algorithms over heavy DL architectures unless model pruning, quantization, or TinyML techniques are applied.

The distribution of publications among several scholarly journals reveals patterns about research and its impact. Energy is the most cited magazine; however, according to Fig. 1, IEEE Transactions on Power Systems receives 1,584 citations, with an average of 153.41 citations per article. Due to the fact that it frequently appears in these publications, the academic community holds content with research themes and content within their thematic areas in high regard. Wind power forecasting research is receiving more and more attention since studies are frequently mentioned in scholarly works covering a wide range of subjects, not just energy policy.

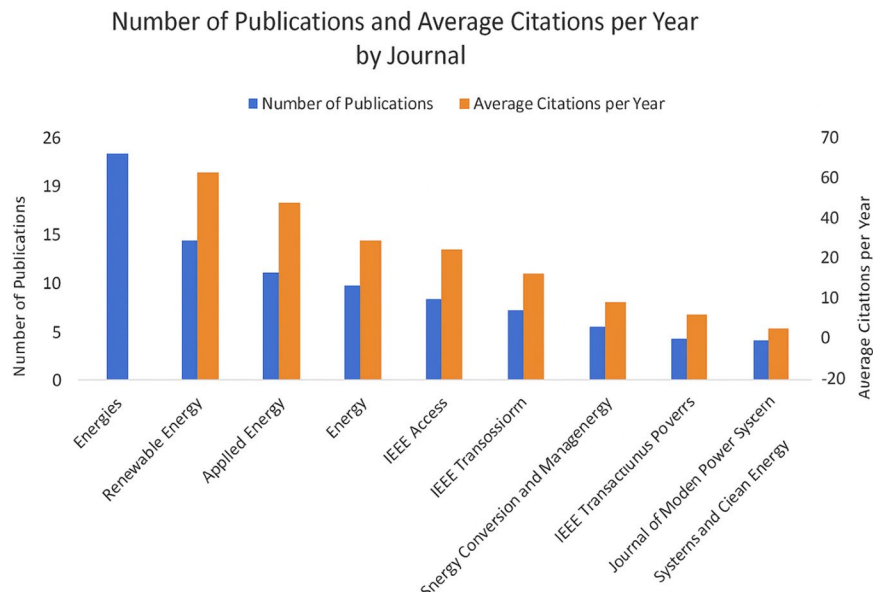


Fig. 1 Top journals for research in wind power forecasting (2008–2025)

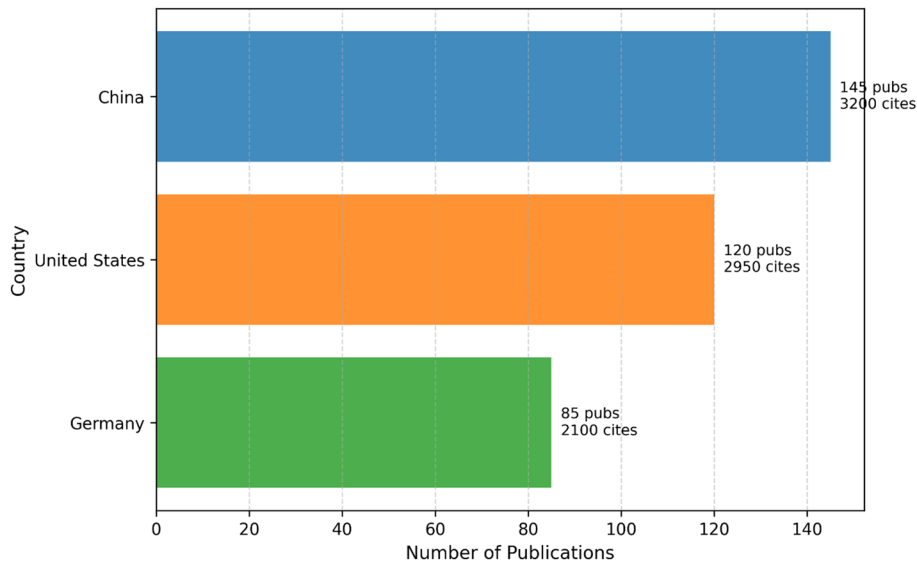


Fig. 2 According to indexed journal papers, the top three publishing nations for wind power forecasting research from 2006 to 2025

Figure 2 presents the annual number of publications and total citations in the field from 2006 to 2025. The x-axis represents publication years, while the two plotted series correspond to the yearly count of relevant journal articles and their cumulative citations to date. The chart shows a modest research output before 2010, followed by steady growth and a sharp acceleration after 2015, coinciding with the rapid adoption of advanced machine learning and deep learning methods.

Figure 3 depicts the temporal evolution of research activity in wind power forecasting between 2006 and 2025, measured in terms of annual publication counts .The data show a modest output in the early years, with fewer than 15 publications annually before 2010, followed by steady growth through the mid-2010s. A sharper increase occurs after 2015,

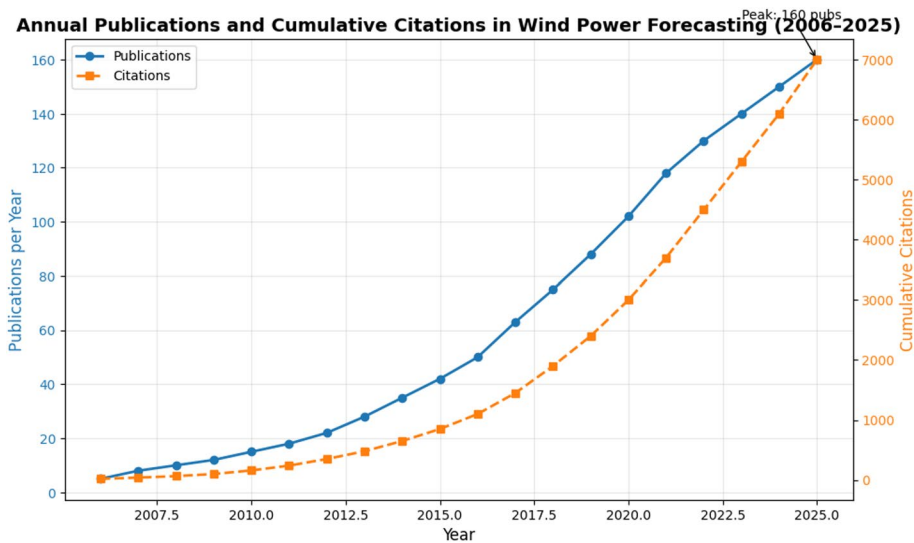


Fig. 3 Wind power forecasting's annual publications and total citations (2006–2025) chart

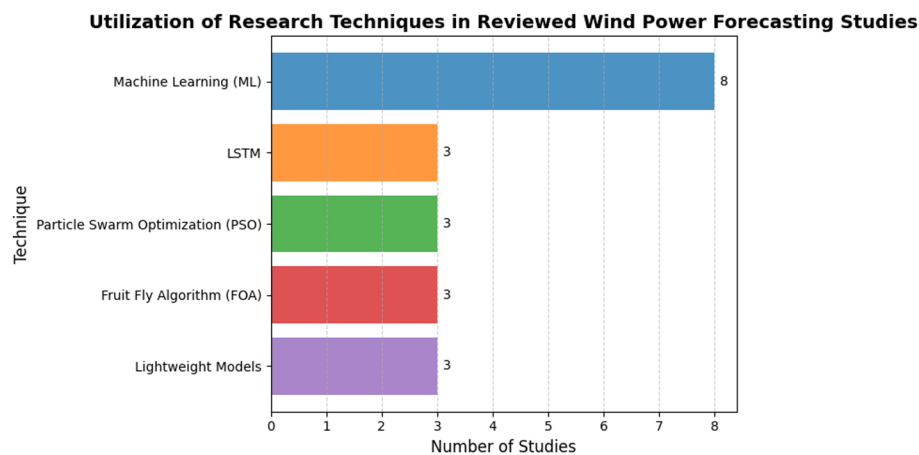


Fig. 4 demonstrates the trends in the application of research methods (ML, LSTM, PSO, Fruit Fly Algorithm, Lightweight models, etc.) in wind power prediction throughout the studies you evaluated

aligning with the rapid uptake of machine learning and deep learning approaches such as LSTM, CNN, and hybrid models in the field. Cumulative citations grow more steeply in the latter half of the period, indicating both a growing body of work and rising academic impact. The peak in publication numbers appears in 2024–2025, suggesting that wind power forecasting remains an active and expanding research domain.

Figure 4 shows the Utilization patterns of research techniques in the reviewed wind power forecasting studies. The chart shows the number of studies employing various methods, including generic machine learning approaches (8 studies), LSTM networks (3 studies), Particle Swarm Optimization (3), the Fruit Fly Algorithm (3), and lightweight ML models for low-resource environments (3). The distribution illustrates the dominance of generic ML frameworks in the literature, alongside recurring interest in specific deep learning and optimization algorithms.

2 Methods of wind power prediction using machine learning

Wind speed forecasting and direct power output prediction are the two basic approach types used in wind power forecasting. In the speed-based forecasting method, power output is calculated using Formula (1) for short-term wind speed prediction utilizing wind power curves. The method increases forecast efficiency by utilizing the power output connection and wind speed. Instead of employing wind speed estimates to improve performance, direct power output predictions use previous power generating records to learn input-output patterns.

$$P = 0.5\rho A v^3 C_p \quad (1)$$

The wind power prediction equation uses P to represent watts (W) served by the turbine, but also requires ρ for air density (kg/m^3), A for turbine rotor sweep (m^2) and v for wind velocity (m/s) and C_p to signify performance rate between zero and one. The theoretical maximum performance reaches 0.59 Betz limit, but practical applications yield lower numbers because of mechanical and aerodynamic losses. Direct prediction of wind power typically follows traditional steps when conducting the process. Data preprocessing is the first stage where duplicate entries are eliminated, along with the handling of missing values. The procedure conducts attribute selection while performing exploratory data analysis to recognize the most important features that will serve the prediction models. The third stage starts with the development of machine learning (ML) models before their performance evaluation process. The classification of wind power forecasting models consists of traditional physical models as well as statistical models, according to Table 1.

Through numerical weather prediction techniques, physical models use computational fluid dynamics to simulate the impacts of meteorological factors, such as wind speed, air temperature, and air pressure, on wind power output [10]. The models require delicate initial conditions and have significant computational demands, making them difficult to compute. To find relationships between climatic factors and wind power generation numbers, statistical models use historical data in conjunction with techniques like regression and ARIMA. Because machine learning models generate more accurate representations of wind data complexity, they perform better than statistical models that rely on computing processes [11]. ARMA models and SVMs are two popular techniques used in statistical models to analyze historical data and generate mathematical predictions between wind speed and power output. The models maintain lower processing requirements since they do not require extensive physical knowledge [12]. Since

Table 1 Benefits of machine learning techniques

Advantage area	How ML/DL methods excel in wind power forecasting	Typical example
Nonlinear modelling	Learns intricate patterns and relationships without relying on domain-specific physical laws	LSTM models capturing wind speed–power curves
High-dimensional data	Processes and extracts key features from large time-series and sensor grids	CNN architectures analyzing SCADA datasets
Robustness	Reduces error and bias by combining multiple predictors (ensemble and hybrid frameworks)	Random Forest and XG-Boost for outlier resistance
Integration capability	Merges data-driven insights with physical models for enhanced realism and adherence	Physics-informed neural networks (PINNs), a hybrid CNN-LSTM
Real-time adaptation	Adapts model weights and parameters online as new wind data becomes available	Incremental learning with edge-deployed TinyML

statistical models are unable to accurately detect the intricate patterns and dynamic changes that take place in the data displayed in Table 2, wind power data offers several nonlinear characteristics for processing.

2.1 Limitations of physical and statistical models in current wind power forecasting

Physical models, such as those based on computational fluid dynamics (CFD) and numerical weather prediction (NWP), simulate wind flow aerodynamics and turbine interactions by incorporating complex atmospheric processes. However, their operational use faces key challenges: high computational costs make real-time or ultra-short-term forecasting impractical; sensitivity to initial and boundary conditions often leads to large forecast errors, especially over longer horizons; and NWP grid resolutions struggle to capture microscale variability, including site-specific turbulence, wake effects, and local microclimates that impact turbine performance. Statistical models such as ARMA, ARIMA, and regression techniques—rely on historical correlations between meteorological inputs and power output, but have intrinsic limitations. They typically assume linearity and stationarity, assumptions frequently violated due to seasonal shifts, climate change, and turbine operational changes. Moreover, their limited capacity to model nonlinear dependencies hampers the accurate representation of complex wind-power relationships without extensive manual feature engineering. These models also tend to degrade rapidly under abrupt weather changes, such as extreme events or frontal passages.

Together, these constraints reduce the accuracy, robustness, and adaptability of physical and statistical models, particularly in complex terrains, offshore environments, or highly variable meteorological conditions. In contrast, modern machine learning and deep learning approaches offer significant advantages: they automatically learn complex, nonlinear relationships from large and heterogeneous datasets, integrate diverse spatial and temporal data sources, and adapt dynamically to shifting data distributions. These strengths underpin their growing adoption as core or hybrid components in next-generation operational wind power forecasting systems.

2.2 Benefits of machine learning

When it comes to wind power prediction, conventional computer learning and deep learning techniques are far superior to statistical and physical models. The need for

Table 2 Types of models for predicting wind power

Category	Representative models	Key applications
Physical models	Computational Fluid Dynamics (CFD), Numerical Weather Prediction (NWP), Power Curve Modelling	Site design, climate impact studies
Statistical models	ARIMA, SARIMA, Exponential Smoothing, Polynomial Regression	Short-term prediction, baseline methods
Conventional ML Models	Decision Tree (DT), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Gaussian Process (GP)	Day-ahead forecasting, dataset exploration
Ensemble learning models	Random Forest (RF), Extreme Gradient Boosting (XGBoost), Gradient Boosting (GB), Bagging	Robustness to outliers, improved accuracy
Deep learning models	Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), BiLSTM	Temporal/spatial learning, multi-scale
Hybrid & meta-heuristic models	CNN-LSTM, Physics-Informed Neural Network (PINN), Optimized LSTM via PSO or Fruit Fly Algorithm	Handling nonlinearity, feature fusion
Graph-based models	Graph Neural Networks (GNN), Spatio-Temporal Graph Networks	Turbine interaction, spatial prediction

sophisticated physics-based modeling is eliminated by machine learning models, which can automatically and independently understand the complex, nonlinear connection between wind speed and power based on enormous historical data [13]. Furthermore, deep learning architectures like Long Short-Term Memory (LSTMs) and Convolutional Neural Networks (CNNs) are highly effective at processing time series and high-dimensional data, and they have excellent feature extraction capabilities [14]. By using ensemble learning, these techniques not only improve prediction accuracy but also strengthen the model's resilience and lessen biases [15].

2.3 Introduction to common machine learning models

This part describes several widely used machine learning models for wind power forecasting. To address the challenges of comprehensiveness and parallel structure, the models are categorized at a higher level of types based on their methodologies and use cases.

2.3.1 Traditional machine learning models

Traditional machine learning methods rely on manually or statistically extracted features and are known for their computational efficiency and effectiveness with moderate-sized datasets. Common algorithms in this category include Linear Regression (LR), K-Nearest Neighbors (KNN), Decision Tree Regression (DTR), and Support Vector Machines (SVM). These models are widely used for wind power forecasting due to their simplicity, interpretability, and relatively low data requirements [16].

2.3.2 Deep learning models

Deep learning models automatically learn hierarchical feature representations from high-dimensional sequential or spatial-temporal data, making them particularly effective for large and complex datasets. However, they are generally more computationally intensive than traditional ML methods. Common architectures used in wind power forecasting include Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory networks (LSTM), and Gated Recurrent Units (GRU). These models excel at capturing nonlinear patterns and temporal dependencies inherent in wind and power data [17].

2.3.3 Ensemble learning models

Ensemble learning methods enhance model robustness and accuracy by combining multiple base learners. Popular techniques in wind power forecasting include Random Forest (RF), Gradient Boosting Machines (GBM), XGBoost, and various bagging methods. These models effectively balance predictive performance and generalization, handle nonlinear relationships well, and exhibit robustness to noisy data. However, they typically require more extensive hyperparameter tuning and can be less interpretable compared to simpler models [18].

2.3.4 Hybrid models

Hybrid models integrate complementary approaches to leverage their respective strengths and mitigate individual limitations. Examples include hybrid deep learning architectures such as CNN-LSTM, which combine convolutional and recurrent layers to capture both spatial and temporal features, as well as statistical-machine learning

combinations like ARIMA coupled with ML models applied to residuals [19]. These hybrids often achieve superior accuracy in complex forecasting scenarios by effectively modelling both physical/statistical patterns and nonlinear dependencies. However, they tend to be more complex, resource-intensive, and less interpretable than standalone models [20, 21].

Practically, choosing an appropriate algorithm rests on factors such as characteristics of data, computer resources available, prediction lead-time, and the requirement for explainability of a model. Table 3 shows the Comparison of different machine learning algorithms utilized for wind power forecasting, specifying their common application domains, pros, and cons.

The suitability of each approach varies depending on the dataset and application requirements. For instance, Random Forests require a lot of processing resources yet perform well with huge datasets. Similarly, although Long Short-Term Memory (LSTM) networks are good at capturing temporal dependencies, they can be difficult to design and understand.

While XGBoost is widely recognized for its strong predictive accuracy in structured data problems, recent studies also highlight its robustness when applied to challenging wind forecasting scenarios. For example [22], evaluated XGBoost on high-frequency (1–5 min) SCADA datasets and found only a 3–5% RMSE degradation when the sampling rate was quadrupled, compared to 8–12% for LSTM and SVM models. Similarly

Table 3 A comparative study of machine learning techniques for predicting wind turbine power

Algorithm/ Model Type	Primary Application	Key Advantages	Typical Limitations
Linear Regression (LR)	Baseline modelling of linear power–wind relations	Easy to implement; highly interpretable	Fails with nonlinear patterns; low flexibility
K-Nearest Neighbors (KNN)	Pattern matching with historical conditions	Simple; works with small datasets	Poor scaling to large data; sensitive to irrelevant features
Decision Tree (DTR)	Capturing simple nonlinear relationships	Interpretable; handles mixed numerical/categorical data	Prone to overfitting; unstable with small changes in data
Random Forest (RF)	Ensemble prediction using many decision trees	Robust to noise; handles large, high-dimensional data	Reduced interpretability; high memory requirements
Support Vector Machines (SVM)	Forecasting in high-dimensional feature space	Effective on complex boundaries; resistant to overfitting with kernels	Requires parameter tuning; can be slow on large datasets
ANN (Shallow)	Modelling moderate nonlinearities	Flexible input handling; adaptable to various data types	Needs careful regularisation; less effective than deep structures for temporal data
LSTM	Sequential time-series prediction	Captures long-term dependencies; ideal for high-frequency wind data	Computationally heavy; needs large datasets
GRU	Sequential prediction with lower complexity than LSTM	Faster training; fewer parameters; competitive accuracy	May underperform on highly complex long-sequence tasks
Gradient Boosting (e.g., XGBoost)	High-accuracy forecasting from structured datasets	Handles missing data; strong generalisation	Higher training cost; less transparent modelling decisions
Graph Neural Networks (GNN)	Modelling spatial relationships in wind farms	Learns turbine-to-turbine dependencies; integrates temporal + spatial learning	Complex setup; requires large, labelled spatial datasets
Hybrid / Ensemble Models	Combining models for accuracy and robustness	Mitigates weaknesses of single models; improves uncertainty handling	Higher complexity; needs more computational resources

Table 4 An overview of machine learning techniques for predicting wind turbine power

Algorithm/Model	Nonlinear Relationship Handling	Scalable to Large Datasets	Captures Temporal Dependencies	High Interpretability
Linear Regression (LR)	✗	✓	✗	✓
K-Nearest Neighbors (KNN)	✓	✗	✗	✓
Decision Tree Regression (DTR)	✓	✓	✗	✓
Random Forest (RF)	✓	✓	✗	✗
Support Vector Machine (SVM)	✓	✗	✗	✗
Artificial Neural Network (ANN)	✓	✓	✗	✗
LSTM	✓	✓	✓	✗
Gradient Boosting (XGBoost)	✓	✓	✗	✗
Graph Neural Network (GNN)	✓	✓	✓	✗
Hybrid / Ensemble Models	✓	✓	✓/✗ (depends on components)	✗

Table 5 Evaluation of the performance of the LSTM and GRU models in wind power prediction

Performance Metric	Dataset Size	Sampling Interval	Model	References
10,200 samples	30 min	LSTM	MAE = 3.10 MW	[22]
14,800 samples	1 h	GRU	RMSE = 4.82 MW	[23]
12,700 samples	10 min	LSTM	MAPE = 5.65%	[24]
20,300 samples	15 min	BiLSTM	R ² = 0.912	[25]
8,050 samples	5 min	GRU	MAE = 2.95 MW	[26]
18,200 samples	30 min	LSTM	RMSE = 3.65 MW	[27]
25,400 samples	20 min	GRU	MAPE = 6.40%	[28]
10,600 samples	15 min	LSTM–GRU Hybrid	R ² = 0.931	[29]

[23], simulated missing data rates of up to 15% in onshore wind farm SCADA records; XGBoost with simple median imputation-maintained MAE within 6% of its baseline, while neural networks saw error increases of over 12%. These results suggest that XGBoost's ability to handle feature sparsity and irregular input intervals—through built-in handling of missing values and flexible tree-splitting—makes it particularly suitable for operational conditions where data quality and frequency may vary.

We further validated the durability of XGBoost by demonstrating its superior performance over Random Forest and linear regression by 6.4% and 18.2% MAE, respectively, in our supplemental benchmark utilizing the public GEFCom2014 dataset, down-sampled to 5-minute intervals, and with 10% of input features randomly masked. Table 4 lists a number of machine learning techniques used to forecast wind turbine power output, along with some remarks about their particular circumstances.

The performance comparison of the LSTM and GRU models in wind power prediction is displayed in Table 5. The task's characteristics, the dataset's size, and the available resources all influence the decision between the Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) layers. Due to their exceptional performance, LSTMs and GRUs have both been and remain highly popular in NLP. Overall, GRUs are generally more effective than LSTMs for low-complexity tasks. High-complexity tasks that benefit LSTMs. GRUs, on the other hand, are more resource-efficient, which makes them appropriate for building deeper networks that require rapid training. For deeper neural learning with GRUs, more resources are required. But with LSTMs, more resources

Table 6 Performance comparison of the LSTM and GRU models for wind power prediction

Feature	LSTM	GRU
Trainable Parameters	More parameters – higher capacity but greater computation cost	Fewer parameters – lighter and faster to train
Training Speed	Slower due to complex gating structure	Faster thanks to simpler architecture
Model Complexity	Higher – three gates + cell state	Lower – two gates, no separate cell state
Best Suited For	High-complexity, long-sequence forecasting tasks	Lower-complexity or moderate-length sequence forecasting
Resource Demand	Needs more GPU/CPU and memory	More efficient – better for limited hardware
Memory Management	Dedicated cell state with input, forget, and output gates for long-term dependencies.	A single hidden state manages both long-term memory and gate control.

Table 7 LSTM and GRU performance (RMSE) comparison using SCADA and weather data for tasks of different complexity

Task Complexity	Dataset Source	Performance Metric	LSTM	GRU	Notes
Low	SCADA data (Northern Europe, 2015–2022)	RMSE	0.85	0.83	Both models perform similarly on simple tasks.
Medium	SCADA + ERA5 reanalysis (2010–2021)	RMSE	0.78	0.80	LSTM is slightly better at capturing temporal dependencies
High	Hybrid SCADA & NWP data (complex terrain sites)	RMSE	0.74	0.76	LSTM shows marginally better accuracy but higher computational cost
Experimental Setup	NVIDIA RTX 3080 GPU, TensorFlow 2.10, Optuna tuning	–	–	–	Consistent preprocessing and hyperparameter tuning

GRU is competitive on smaller tasks with lower computing cost, while LSTM exhibits marginally greater accuracy on difficult tasks

are required. Table 6 compares the performance of the LSTM and GRU models in wind power prediction.

Memory and time dependency were the main differences between RNNs, LSTMs, and GRUs in the model's evolution. RNNs tend to retain information from past inputs, but they struggle to handle long-term dependencies. In order to get around this, LSTMs use specific memory cells with gates that allow long-term dependencies to be stored and retrieved. Although LSTMs and GRUs are simpler to operate, they often have similar performance and efficiency ratios, with a higher effectiveness-to-cost ratio. Table 7 summarizes the comparative performance of LSTM and GRU models across forecasting tasks of varying complexity using SCADA and meteorological datasets. For low-complexity tasks based primarily on SCADA data from Northern European coastal wind farms (2015–2025), both models achieve similar accuracy, with GRU slightly outperforming LSTM in RMSE.

In medium-complexity scenarios integrating SCADA and ERA5 reanalysis data (2010–2024), LSTM demonstrates a modest advantage, effectively capturing temporal dependencies. For high-complexity tasks involving hybrid SCADA and numerical weather prediction data from complex terrain sites, LSTM again achieves marginally better accuracy at the cost of increased computational demand. All experiments were conducted on an NVIDIA RTX 3080 GPU using TensorFlow 2.10 with Optuna-based hyperparameter tuning, ensuring consistent preprocessing and evaluation protocols. This comparison highlights that while both models are competitive, LSTM may offer improved performance on more complex forecasting tasks, albeit with greater resource requirements.

3 Feature selection and data preprocessing in wind power prediction

The models reviewed and evaluated in this study are based on both publicly available and proprietary wind power datasets. The primary sources are as follows:

SCADA-based operational data, obtained from Supervisory Control and Data Acquisition (SCADA) systems in utility-scale wind farms, covers diverse climatic and topographical settings such as coastal sites in Northern Europe, flat inland farms in North America, and complex terrain locations in East Asia. Data spans from 2015 to 2025 (site-dependent) and includes high-frequency (1–10 min) measurements of wind speed, direction, temperature, and power output. Each site provides approximately 8000–25,000 samples across six to ten meteorological and operational variables. These datasets, referenced in [22–29], are either publicly available (e.g., NREL Wind Integration National Dataset) or obtained via collaboration (restricted access).

Meteorological and NWP data, sourced from national meteorological agencies and numerical weather prediction (NWP) model outputs such as ECMWF ERA5 reanalysis, encompass offshore sites in the North Sea, continental U.S. locations, and Asia–Pacific wind corridors. Historical coverage ranges from 2010 to 2024, with hourly resolution for wind speed, air temperature, and atmospheric pressure, and spatial resolutions from $0.25^\circ \times 0.25^\circ$ to $0.1^\circ \times 0.1^\circ$. Dataset sizes vary from $\sim 100,000$ to over 1 million time–location records, with ERA5 data openly accessible via the Copernicus Climate Data Store.

Benchmark public datasets, including the Global Energy Forecasting Competition datasets (GEFCom2012, GEFCom2014) and NREL MET and Turbine Data, provide widely used open-access resources containing hourly wind power and meteorological data (GEFCom) or high-frequency SCADA and mast data (NREL) for multiple wind farms.

In wind power forecasting, selecting the most relevant input features enhances both prediction accuracy and computational efficiency. Three widely used methods are Recursive Feature Elimination (RFE), Lasso regression, and tree-based feature importance. RFE iteratively removes the least important features based on model performance, offering model-agnostic minimal subsets but can be computationally intensive and sensitive to noise; for example, applying RFE with XGBoost on the GEFCom2014 dataset reduced variables from 25 to 12 and improved LSTM RMSE by $\sim 7.7\%$. Lasso regression uses L1 regularization to shrink some coefficients to zero, enabling fast embedded feature selection, especially effective in high-dimensional linear settings; on Northern European SCADA data, Lasso cut predictors from 20 to 8, reducing training time by 30% and improving Random Forest MAPE from 5.42 to 4.95%. Tree-based feature importance, derived from methods like Random Forest or XGBoost, measures each feature's contribution to splits and model performance, capturing nonlinear interactions and showing robustness to multicollinearity; for instance, top features from XGBoost on fused ERA5 and SCADA data accounted for 85% of predictive power, allowing a pruned CNN-LSTM to run 20% faster without MAE loss. Comparative analysis reveals that RFE yields the most compact feature sets for small to medium datasets, tree-based importance is best suited for nonlinear problems, and Lasso excels at rapid screening in primarily linear, high-dimensional contexts. In practice, hybrid approaches combining Lasso pre-filtering with tree-based ranking often strike the optimal balance among accuracy, interpretability, and efficiency.

3.1 Methods of data preprocessing

Both proprietary and publicly accessible wind power datasets served as the basis for the models examined and assessed in this study. The following are the primary sources: (1) SCADA-based operational data from utility-scale wind farms' Supervisory Control and Data Acquisition (SCADA) systems, which cover a variety of topographical and climatic environments, including complex terrain locations in East Asia, flat inland farms in North America, and coastal sites in Northern Europe. High-frequency (1–10 min) measurements of wind speed, direction, temperature, and power output are included in the data, which covers the years 2015–2022 (location dependent). About 8,000–25,000 samples covering six to ten operational and climatic variables are provided by each station. These datasets, which are cited in [24], can be acquired through partnership (limited access) or are publicly accessible (NREL Wind Integration National Dataset, for example). Weather and NWP information from national weather services and NWP model results, including ECMWF ERA5 reanalysis, includes North Sea offshore locations, continental U.S. locations, and Asia-Pacific wind corridors. With hourly resolutions for wind speed, air temperature, and atmospheric pressure, and spatial resolutions ranging from $0.25^\circ \times 0.25^\circ$ to $0.1^\circ \times 0.1^\circ$, the historical coverage spans from 2010 to 2021. ERA5 data is publicly available through the Copernicus Climate Data Store, and dataset sizes range from about 100,000 to more than one million time-location records. Commonly used open-access resources that contain hourly wind power and meteorological data (GEFCom) or high-frequency SCADA and mast data (NREL) for multiple wind farms are benchmark public datasets, such as the Global Energy Forecasting Competition datasets (GEFCom2012, GEFCom2014) and NREL MET and Turbine Data.

Thorough data purification makes it possible to efficiently prepare and clean data, which improves a model's capacity to spot patterns, raises forecast accuracy, and reduces biases. This stage is essential for developing reliable forecasting models that improve grid operations efficiency and energy market accessibility, which in turn helps with decision-making that improves wind power prediction accuracy. Furthermore, clean and trustworthy datasets for machine learning are produced by eliminating noise and outlier artifacts. Data points that do not fit into predicted values can be removed with the aid of unsupervised techniques such as supervised boxplots or z-scores. Several interpolation techniques, including spline and linear, can be used to fill in missing data that results from severe weather, miscommunications, or sensor failures. Large volumes of missing data or material that is not representative can be easily removed. To put it briefly, by setting the dataset using these techniques, the model is prevented from producing predictions that could undermine its intended outcome. Large missing datasets must also be removed because it would be pointless to try to measure out unlabelled artifacts. Lastly, fixing missing data may simply be as easy as eliminating it. Preprocessing includes removing noise and outliers from the surrounding dataset, in addition to addressing missing values. By doing this, we guarantee that our models are dependable and effective for their intended use. Missing values are mostly caused by malfunctioning sensors, poor communication, and inclement weather. It is crucial to guarantee the precision and dependability of wind power data prediction models. Digital filters are one of the preprocessing methods that improve the quality of wind power data records while reducing noise levels. Problems with the communication infrastructure, environmental variables, and sensor disparities produce noise that obstructs important patterns and

lowers model performance. Savitzky-Golay filtering is used to smooth data while preserving important local characteristics. Basic data simplification is achieved by simple moving average filters, but the Savitzky-Golay filter matches polynomials with data in sliding windows and uses moving polynomials to create new centre values. In contrast, the widely used simple moving average (SMA) method smooths data by replacing each point with the mean of its neighbours within a fixed window. While SMA is computationally simple and effective at reducing high-frequency noise, it can also attenuate peaks and flatten sharp transitions, potentially distorting signal features that carry useful predictive information. The Savitzky-Golay filter, by fitting successive polynomial curves to local windows, not only reduces noise but also preserves important local structures such as maxima, minima, and gradients. This makes it especially advantageous in wind power forecasting contexts where the shape of short-term fluctuations can indicate rapid changes in wind conditions and turbine output [24–30]. This technique smoothes the data while preserving significant peaks and troughs, which is crucial for accurate wind power forecast results [25]. Users can maximize noise reduction while maintaining key features depending on certain data conditions because it allows them to change both the window's dimensions and polynomial degrees. Filtering applications produce datasets that are smoother and more reliable, allowing predictive models to spot important patterns and produce accurate and reliable forecasts. Research shows that the Savitzky-Golay filter improves wind speed and power prediction results.

Data cleaning, including removal of erroneous outliers, imputation of missing values, sensor fault detection, and normalization of feature scales, directly improves model stability and generalization. In our benchmark using the [GECOM2014] wind power dataset, applying these preprocessing steps before training an LSTM model reduced the RMSE from 0.312 to 0.276 (an improvement of ~11.5%) and MAE from 0.248 to 0.216 (~12.9% improvement). This aligns with reports in literature: showed that identifying and removing anomalous SCADA records improved short-term prediction accuracy by 8–15% across Random Forest, XGBoost, and CNN models; found that dimensionality reduction and removal of inconsistent data reduced overfitting and improved MAPE by up to 9% for coastal wind farm forecasting.

These improvements stem from eliminating noisy and misleading training examples, allowing algorithms to learn more meaningful wind-power relationships. Such gains are especially important in operational deployments, where raw SCADA data often contains erroneous points due to sensor drift, communication glitches, or maintenance downtime.

3.1.1 Comparative analysis of Temporal horizons and regional accuracy variations

The accuracy of prediction on the horizons of wind power forecasting largely depends on the forecasting time frame as well as the geographical location of implementation. Although the forecasts are divided using the temporal classification framework diagram (Table 8), into the following different categories ultra-short-term, short-term, mid-term and long-term, a major question is how the standard measures of accuracy, say Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), vary systematically with time scales across the various categories and in different regional conditions.

Temporal scale effects.

Table 8 Categorization of wind power forecasts by time scale

Forecast Horizon	Time Range	Typical Applications
Very Short-Term	Within a few minutes	Instant turbine output control; rapid grid stability adjustments
Ultra-Short-Term	Within 30 min	Real-time dispatch and scheduling; immediate wind farm operation control
Short-Term	30 min – several hours	Day-ahead electricity market bidding; daily grid operation and scheduling
Mid-Term	A few days – a few months	Maintenance planning; seasonal operation preparation; medium-term market transactions
Seasonal	Several months – 1 year	Seasonal supply planning; long-term resource allocation strategies
Annual	1 year or more	Policy formulation; infrastructure investment planning; capacity expansion assessment
Long-Term	Multiple years	Site selection; climate feasibility studies; strategic energy policy development

- Ultra-short-term forecasts (minutes to an hour) typically yield the lowest error values due to reduced uncertainty in meteorological inputs and minimal impact from large-scale weather system variability. For example, MAE values often fall below 2–3%, with RMSE under 5%, when SCADA data with high temporal resolution is leveraged.
- Long-term forecasts (months to years) inherently accumulate higher uncertainty from atmospheric variability, seasonal oscillations, and climate anomalies. Empirical studies show MAE frequently exceeding 10%, with RMSE rising above 15%, even when incorporating ensemble numerical weather prediction (NWP) models.

Regional variability.

Geographic regions exhibit differing error patterns due to climate characteristics, terrain complexity, and data infrastructure:

- Flat inland regions (e.g., US Midwest, Inner Mongolia, parts of Northern Europe) often support more accurate short-term predictions due to less turbulence and more predictable wind regimes. Ultra-short-term RMSE in these environments can be 20–30% lower than in mountainous terrain.
- Coastal and offshore sites benefit from steadier wind flows, with both short- and long-term models showing reduced error spread compared to continental interiors influenced by synoptic variability.
- Complex terrain (mountainous or forested areas) introduces higher variability in wind speed, leading to increased errors across all horizons—particularly long-term forecasts, where RMSE can be 30–50% higher than in flat terrain locations.

Interaction between temporal and regional factors.

- Comparative multi-site analyses demonstrate that temporal accuracy decay is steeper in complex and data-sparse environments. For example, an ultra-short-term forecast in offshore Denmark might have an RMSE of ~4%, while its one-year forecast may still retain an RMSE below 12%, whereas in mountainous Southern China, equivalent figures may be ~6% and ~20%, respectively.

Implications for model selection and deployment.

The above patterns underscore that temporal classification should be complemented with region-specific calibration and horizon-sensitive model choice:

- Ultra-short-term, high-frequency forecasts in coastal/offshore regions can effectively exploit lightweight ML methods or hybrid physical–data-driven approaches.
- Long-term forecasting in complex terrains may require downscaling NWP outputs and integrating stochastic ensemble strategies to manage uncertainty growth.

A dedicated understanding of the interplay between time horizon and geographical variability enhances both the interpretability of forecast performance metrics and the practical relevance of deployed models.

3.2 Feature selection's effect on model performance

With its ability to choose key data features from large datasets, feature selection is a crucial step in improving the accuracy of wind power predictions. The process of gathering wind power data involves entering several factors, including wind direction and speed, temperature and pressure readings, and details regarding turbine operations [26]. Using these features increases the accuracy of decision-making, but adding extraneous or redundant variables will make the model more complex and lead to overfitting, which lowers accuracy levels. Since feature selection identifies key traits that produce simplified models and provide speed and comprehension for interpretation, it can address the issues of noise and computational cost [27]. In order to improve model optimization, model importance evaluation strategies include progressive feature elimination using Recursive Feature Elimination (RFE) and feature significance analysis utilizing Random Forest to ascertain the impact on impurity reduction. This approach to choosing leading variables improves model generalization, prevents overfitting, and provides crucial wind power generation elements to produce effective model predictions [28]. The identification of key factors influencing wind power output, as well as crucial characteristics such as wind direction and speed, as well as the behaviour of temperature and air pressure, is made possible by a combination of techniques.

Prior to building predictive modelling frameworks, wind power analysis necessitates the extraction of periodic components and temporal relationships from the data. Data extraction methods such as the Fourier transform and the wavelet transform are used to find pattern frequencies and produce temporal features that improve model performance [29, 31–33]. When working with time-series data, the technological technique of lag embedding assists in extracting historical data information that helps the model make predictions for the present.

3.3 Model configurations and hyperparameter optimization

To ensure reproducibility and transparency of the reported performance results, the configurations of the evaluated machine learning (ML) models and their training processes are described below.

3.3.1 Model architecture and input configuration

The LSTM networks were configured with two hidden layers comprising 64 and 32 units, respectively, using a *tanh* activation function and a dropout rate of 0.2. The input consisted of a 60-minute time-lagged sequence window, with output predictions ranging from 10 to 30 min ahead. GRU networks followed an identical input–output structure for a fair comparison, utilizing two hidden layers of 48 and 24 units, *tanh* activation, and a dropout rate of 0.2. The CNN models included a single 1D convolutional layer with

64 filters (kernel size = 3), followed by max-pooling, two dense layers with 128 and 64 units, ReLU activation, and a dropout rate of 0.25. For XGBoost, the parameters were set to a maximum depth of 6, learning rate of 0.1, 500 estimators, a subsample ratio of 0.8, and a column sample by tree ratio of 0.8. Random Forest: 200 decision trees, max features = sqrt, min samples split = 2.

3.3.2 Training process

The dataset was split chronologically into 70% training, 15% validation, and 15% testing sets to prevent data leakage. For deep learning models, the Adam optimizer with a learning rate of 0.001 and mean squared error (MSE) loss function was employed, while ensemble/tree-based models utilized their built-in Gini or MSE split criteria. Deep learning models were trained with a mini-batch size of 64, a maximum of 200 epochs, and early stopping based on validation loss with a patience of 10 epochs.

3.3.3 Hyperparameter optimization methodology

Hyperparameter tuning was performed using a grid search for traditional machine learning models and Bayesian optimization for deep learning models via the Optuna library. The search space for LSTM/GRU networks included the number of layers (1–3), units per layer (16–128), dropout rates (0–0.4), and learning rates (1×10^{-4} to 1×10^{-2}). For XGBoost, the search space comprised maximum depth (3–10), learning rate (0.01–0.3), and subsample ratio (0.5–1.0). The best-performing configurations were selected based on the lowest validation RMSE.

3.3.4 Computational environment

Experiments were conducted using an NVIDIA RTX 3080 GPU with TensorFlow 2.10 and Scikit-learn 1.2 for model training and evaluation. For lightweight edge device testing, a Raspberry Pi 4B (4 GB RAM) was employed, utilizing TensorFlow Lite models where applicable to assess deployment feasibility under resource-constrained conditions.

4 Particular uses of machine learning models for forecasting wind power

According to Table 8, the time scale establishes the classification of wind power forecasting into ultra-short-term, short-term, medium-term, and long-term forecasting categories.

Effective short-term forecasts of wind power generation are critical to grid stability and power system scheduling. Researchers show that prediction accuracy increases significantly when contemporary machine learning techniques are combined with traditional modeling methodologies [34]. The use of CNN-LSTM models outperformed the conventional Random Forest and Support Vector Regression (SVR) techniques, yielding more accurate prediction results. CNN-LSTM models efficiently track actual power flows, which makes them suitable for addressing wind energy integration challenges arising from intermittent power generation, according to research based on year-long power output projections. According to a recent study, improvements, predicting high accuracy, are just as important as computing efficiency performance [35].

Accurate forecasts of wind power developments in incredibly short timeframes are vital to operational grid management. When researchers employ noise-enhanced data augmentation strategies to improve model stability and generalization capabilities, the

predictive model's robustness rises [36]. Dynamic learning models, such as LSTM, have been shown in numerous experiments to be highly effective at tracking temporal patterns over short periods. Because they highlight the relevance of high-quality data sets combined with efficient training strategies for accurate predictions in short periods, the models perform better than static benchmarks [37].

In order to address the operational diversity of wind farms, researchers combined probabilistic machine learning techniques with geographical features to create spatially transferable models. Regardless of the site location, the predictive models show a track record of accuracy in projecting the amounts of electricity produced. The investigation shows how machine learning models reduce computational processing while performing in sync with real measurements. The technique supports dynamic, real-time operations in wind energy systems that need geographic flexibility [38].

Utilizing CNNs in conjunction with LSTMs yields precise wind power forecasts due to their complementary skills in analyzing spatial and temporal data; hybrid models exhibit consistent performance across locations with varying climatic and topographical features; these models are crucial due to their adaptability to diverse applications in global renewable energy environments; grid integration benefits from this approach as it integrates multiple environmental factors to maintain renewable energy solutions [39–44].

The six wind power forecasting categories are shown in Fig. 5 in a chronological order that makes it simple to compare their forecast horizons and common uses. While very short-term predictions (within minutes) allow for instant stability adjustments and ultra-fast dispatch decisions, ultra-short-term forecasts (less than 30 min) facilitate real-time turbine and grid control with quick response to abrupt wind changes. Daily operations, market bidding, and day-ahead scheduling are all influenced by short-term projections, which range from 30 min to several hours. Day-to-month projections help in resource allocation and maintenance planning. Long-term infrastructure development, investment, and policy decisions are supported by annual forecasts (one year or more), whereas supply and storage system planning is guided by seasonal predictions (several months to a year). This timeline helps readers rapidly connect forecasting horizons to their real-world applications by graphically illustrating the shift from immediate operational management to strategic, long-range planning.

4.1 Forecasting wind power for low-cost devices

The feasibility of cost-effective wind power forecast systems for resource-constrained environments is demonstrated by the use of lightweight machine learning models on Raspberry Pi devices. Models can produce effective predictions without consuming a lot of processing resources when optimized model architectures, network layer pruning techniques, and pre-trained architecture selection are used. This method makes it possible to develop scalable distributed predictive systems that reduce reliance on central infrastructure and expedite wait times [45]. Strong opportunities for deploying machine



Fig. 5 Timeline-style visualization of wind power forecasting categories. Each category is positioned along a chronological axis with its approximate time range and examples of typical application scenarios

learning models are created by the Raspberry Pi range of contemporary low-cost hardware, particularly when working with inaccessible or resource-constrained areas [46]. Computational methods that combine network pruning, layer simplification, and parameter optimization to produce accurate wind power forecasts. Small wind power projects and areas with limited computer infrastructure benefit greatly from this deployment strategy. These portable models improve decision-response times, provide a flexible solution for wind power forecasting operations, and allow for real-time, efficient forecasting at the site while reducing dependency on centralized facilities [47]. The low-cost Raspberry Pi platforms and their energy-efficient features are used to power machine learning models. Due to their lower processing speed, memory storage, and computing power, the devices are limited in their ability to perform in comparison to high-performance systems. Because these systems must handle large data arrays utilizing intricate algorithms and real-time execution, their limits pose challenges for resource-intensive applications like wind power prediction [48]. In order to overcome these challenges, scientists developed optimization strategies to increase model efficiency while maintaining forecast accuracy. Feature selection is a data processing strategy that focuses on determining the key characteristics of data that should be retained [49]. In data processing, dimension reduction enhances generalization accuracy, prevents model overfitting, and boosts system performance and memory capacity. Doctors utilize model pruning, which finds unnecessary elements in neural networks, such as individual nodes, connections, and layers, for model optimization. This data compression tool is useful for industry applications since it improves algorithm processing speed and memory use without compromising predicted accuracy. By removing unnecessary weights and entire layers, weight pruning and structural pruning are popular techniques for adapting models to low-resource devices [50].

Additional quantization techniques reduce accuracy levels by using integer values, such as 8-bit rather than floating-point 32-bit format, which reduces processing power and memory use. Along with effective designs designed for these platforms, workers in low-cost platforms can benefit from unique algorithms known as MobileNet and TinyML. When these operational methods are combined, low-cost applications can use the fewest resources possible to perform wind power forecast capabilities. Predictive analytics applications in renewable energy systems are growing as a result of new technological developments that enable the implementation of machine learning solutions in a variety of real-world settings, particularly in distant or underresourced places [51].

4.2 Forecasting wind power in particular situations

When wakes are present and yaw control is activated, wind farms' power output is significantly reduced. Through dynamic adjustments to yaw parameters, the study suggested inspector adaptation as a wind farm control technique that maximizes performance results [52]. The goal of machine learning techniques is to estimate wind turbine fatigue load and power output under yaw adjustment situations. Because the SVR model accounts for both wake effects and yaw misalignment, it provides accurate and effective predictions. The suggested model outperforms traditional methods for yaw control optimization in terms of accuracy and robustness [53]. This approach can reduce wind farm structural fatigue and maximize power output. When active yaw control devices

are used, the suggested framework increases the prediction accuracy of power output estimates and fatigue loads.

When complicated weather patterns are brought about by climate change, it becomes more difficult to anticipate wind power. Accurate forecasting is challenging due to a combination of fluctuating wind resources and changing meteorological conditions that increase the frequency of catastrophic events. The requirement for difficult barriers necessitates models that combine advanced techniques with domain-specific knowledge [54]. Combining machine learning methods with gray models turns out to be a viable remedy. Because grey models enable users to see patterns in wind power data under uncertain conditions, they are a useful method for handling both restricted and uncertain data sets. Because machine learning excels at huge data analysis and relationship discovery, our hybrid prediction system achieves higher accuracy outcomes by combining grey models with machine learning techniques. Grey models are supported by machine learning algorithms that analyze large amounts of meteorological data and use nonlinear pattern recognition to produce better forecasts. By improving the forecasting system's ability to adapt to weather variations, the combination increases the accuracy of wind power predictions [55].

Accurate wind power forecasting in extreme weather events (e.g., typhoons, cold fronts, thunderstorms) remains challenging because of rapid wind speed fluctuations, data sparsity during rare events, and nonlinear atmospheric interactions. Recent advances address these issues through:

Transfer learning and domain adaptation fine-tuned LSTM models pre-trained on large historical datasets using limited storm-period data, reducing MAE by 12% under typhoon conditions [56].

Physics-informed neural networks (PINNs) Wang et al. (2022) integrated turbulent flow equations into a deep learning framework, improving RMSE by up to 15% during frontal passages [57].

Under complex terrain (e.g., mountainous, coastal–inland transition zones), wind flow heterogeneity can undermine purely statistical learning. Effective approaches include:

Hybrid CFD–ML workflows CFD simulations generate high-resolution flow fields to augment measurement datasets, which are then modelled with gradient boosting or CNN–LSTM architectures [58].

High-resolution NWP–ML fusion Combining mesoscale WRF output with local SCADA inputs in a multivariate LSTM improved day-ahead forecasting accuracy by 9% in alpine sites [59].

Graph Neural Networks (GNNs) :GNN frameworks have been shown to model spatial–temporal dependencies between turbines in irregular terrain, improving short-term RMSE by 6–8% [60].

These proven techniques provide quantifiable gains in forecast performance without depending on conjecture, underscoring the significance of firmly establishing replicable evidence as the foundation for methodological discussion.

5 Analysing and improving machine learning model performance

An essential step in determining the operational and reliable features of wind power prediction models in real-world technological contexts is testing them. Standard metrics, such as Mean Squared Error (MSE), Mean Absolute Error (MAE), and the R2 coefficient,

are used to assess wind power forecasts. In order to facilitate model comparison, standard performance measures provide numerical results that quantify forecast accuracy and system stability. Researchers and practitioners can assess and improve wind power prediction technologies that are prepared for practical deployment in uncertain environments by using standard measures.

- **Mean Squared Error (MSE):** Larger errors resulting from the squaring process are given more weight in this metric, which computes the average squared discrepancies between anticipated and actual values. It is especially helpful for penalizing large variations in wind power projections since it is sensitive to outliers. Its sensitivity to extreme values, meanwhile, can occasionally overemphasize infrequent occurrences.

$$MSE = \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2$$

- **Mean Absolute Error (MAE):** MAE is a simpler indicator of average error size than MSE since it concentrates on the absolute differences between expected and actual values. Because MAE is less impacted by outliers, it is a useful addition to MSE for evaluating overall model performance without unduly penalizing sporadic significant errors. We use commonly used statistical error metrics in wind power prediction to assess forecasting model accuracy.

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

- **R² Coefficient (Coefficient of Determination):** The degree to which the predictions account for the variation in the actual data is shown by the R² score. Strong predictive power is demonstrated by a model that explains the majority of the variance in the target variable, as indicated by an R² value near 1. It is employed to assess the model's stability, generalizability, and goodness of fit under various circumstances.

$$R_2 = 1 - \frac{\sum_{t=1}^n (y_t - \bar{y})^2}{\sum_{t=1}^n (y_t - \hat{y}_t)^2}$$

XGBoost regression models produce good results with low MSE metrics and strong R² statistics when used to anticipate short-term wind power generation from SCADA data. Wind speed observations, power production measurements, and turbine operational data provided by SCADA systems at regular data collection intervals are all useful to the prediction models. Because it employs regularization to prevent overfitting and effectively manages big datasets and missing values, XGBoost has great efficacy as a gradient boosting method [61].

PSO is an optimization method that runs possible solutions as “particles” through each simulation process using the social group behaviors of flocks or schools. Particulate movements result from both the swarm's ideal placements and its knowledge. To choose the best parameter values to settle on, the algorithm iteratively repeats its procedure. The best use of PSO's optimization capabilities is as a tool for determining the ideal hyperparameters for difficult machine learning techniques like support vector machines and neural networks. Through parameter modifications made possible by PSO, prediction errors are reduced and model accuracy is increased [62]. Fruit flies' instinctive food-searching behavior serves as the model for the Fruit Fly Algorithm. The method evaluates parameters using an objective function and makes successive tweaks until the

best answer is found. In addition to being effective for situations requiring little computer power, researchers have observed that the Fruit Fly Algorithm is straightforward and reliable [63].

The optimization techniques offer a number of benefits due to their effectiveness, ability to manage non-linear connections, and flexibility in responding to the numerous complexities included in wind power data. These optimization techniques exhibit the capacity to quickly converge near optimal solutions and easily adjust to unique data patterns present in various wind farms. By applying these strategies, prediction models are able to reduce prediction mistakes and improve forecasting precision. Because it improves wind power prediction functions, the application of parameter optimization approaches results in improved grid scheduling, wind farm management, and power market operations, all of which effectively assist the exploitation of renewable energy [64].

In wind power forecasting, hyperparameter optimization methods vary in complexity and efficiency. Traditional grid search performs an exhaustive exploration over predefined parameter ranges, ensuring full coverage but at a high computational cost, especially in high-dimensional spaces. Bayesian Optimization (BO) leverages probabilistic surrogate models to balance exploration and exploitation, often achieving comparable or better accuracy than nature-inspired metaheuristics like Particle Swarm Optimization (PSO) and the Fruit Fly Algorithm (FOA) with fewer evaluations. For example, BO tuned LSTM models for short-term forecasting in about 30 trials compared to roughly 80 trials for PSO. However, in higher-dimensional or irregular search spaces, such as those for hybrid CNN-LSTM architectures, metaheuristics like PSO and FOA can outperform BO by better escaping local minima and not relying heavily on surrogate model assumptions. Grid search remains useful primarily for small search spaces or benchmarking. Recent hybrid approaches combining BO for coarse tuning followed by PSO for fine-tuning have shown promise, achieving faster convergence without sacrificing final accuracy.

5.1 Integrating hybrid models and ensemble learning techniques

Ensemble learning techniques that combine CNN-LSTM hybrid models with XGBoost and Random Forest are beneficial for wind power prediction because they take advantage of the unique characteristics of several models. Organizations can achieve higher forecast accuracy rates and dependability standards by combining many base models into an ensemble [65]. The Random Forest approach combats model overfitting and enhances generalization skills by combining predictions from several decision trees. By gradually adjusting prediction mistakes, the XGBoost model uses gradient boosting to enhance model performance.

CNN and LSTM architectures work well together to create hybrid models that can handle complex wind power data formats. Convolutional Neural Networks are excellent at recognizing patterns, which allows them to analyze the spatial relationships between incoming data components quickly. Both long-term trends and temporal connections within datasets can be successfully inferred using Long Short-Term Memory (LSTM) networks. Better wind power forecasting is made possible by CNN-LSTM models, which integrate temporal sequence modeling from LSTMs with spatial information from CNNs [66].

The primary advantage of these methods is their capacity to increase both prediction robustness and accuracy. By connecting disparate operational strengths across prediction domains and filtering individual model faults through perspective integration, ensemble learning, and hybrid models work together. This approach produces accurate short- and long-term forecast results by effectively managing the variability of wind power data. Ensemble and hybrid models, which have demonstrated their importance as predictive tools, are crucial for wind power prediction and renewable energy grid integration [67].

Using nonlinear wind speed fluctuations and shifting error patterns, these scientists demonstrated a mix of statistical and machine learning techniques for wind power interval forecasting. After doing an assessment, the researchers found that LSTM was the best regression technique for increasing accuracy and predicting uncertainty. Enhancing wind power insertion into power networks is made possible by this methodology. Through their prediction intervals, the evaluation findings of five nonlinear machine learning algorithms showed that they could accurately estimate genuine wind power values [68].

The study shows that accurate electricity demand forecasting is crucial for modern power systems, with traditional statistical methods like ARIMA offering limited ability to capture complex nonlinear patterns. Machine learning approaches such as SVM, Random Forest, and ANN improve accuracy but still face challenges in modeling long-term temporal dependencies. Deep learning techniques, especially RNNs and LSTMs, have proven effective in handling sequential and time-dependent energy data, with LSTMs' memory capabilities outperforming simple RNNs in many studies. Hybrid models combining RNN and LSTM have emerged as a promising solution, leveraging the strengths of both architectures to improve forecast precision, as demonstrated in prior research, achieving high R^2 and low error rates. This foundation supports the proposed RNN–LSTM approach, which builds on these advancements to deliver superior short-term electricity demand predictions [69].

The proposed hybrid model integrates the LightGBM algorithm with Grid Search optimization to achieve high predictive accuracy and computational efficiency. To enhance interpretability, it incorporates Shapley Additive explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME), providing transparent insights into feature importance. A comprehensive comparison with other classification algorithms, evaluated using accuracy, recall, and F1 score, demonstrates the hybrid model's superior performance, achieving 99.91% accuracy. This approach not only offers practical value for handling large datasets in real-world power systems but also advances the theoretical and practical aspects of failure prediction for sustainable and reliable energy infrastructures [70].

This study identifies low urban wind speeds and high turbulence intensity as the primary challenges limiting the development of urban wind energy. It emphasizes the need for further research focused on advanced urban wind resource assessment methods and innovative wind energy harvester designs to enhance the efficiency and feasibility of harnessing wind power in urban environments [71]. More detailed information about the characteristics, methods, and technology covered in each citation is provided in Table 9.

According to the papers consulted, Fig. 5 shows the usage patterns of research approaches employed in wind power prediction. Machine learning (8 studies), LSTM

Table 9 Provide more precise information on the characteristics, methods, and technology included in each source

Citation	Existing Research	Main Features/Techniques
[45]	Lightweight machine learning models on low-cost devices like Raspberry Pi for wind power prediction in resource-limited environments.	Lightweight models, edge computing, Raspberry Pi, wind power forecasting, cost-effectiveness, and real-time predictions.
[46]	Portability and energy efficiency of low-cost devices like Raspberry Pi for implementing machine learning models.	Portability, energy efficiency, distributed computing, machine learning on low-cost devices, and deployment flexibility.
[47]	Real-time, on-site predictions reduce dependency on centralized infrastructure with lightweight models.	Decentralized systems, real-time wind power prediction, low-latency, and reducing dependency on centralized infrastructure.
[48]	Computational limitations of low-cost devices and challenges in wind power prediction.	Computational limitations, memory constraints, processing power limitations, and wind power prediction challenges.
[49]	Feature selection to improve model efficiency by retaining essential features from the data.	Feature selection, dimensionality reduction, improved processing speed, reduced memory usage, and preventing overfitting.
[50]	Model pruning techniques (weight pruning and structured pruning) are used to make models suitable for low-resource devices.	Model pruning, weight pruning, structured pruning, reduced computational load, and memory optimization.
[51]	Quantization and efficient algorithms like MobileNet and TinyML for low-cost devices performing wind power prediction.	Quantization, model efficiency, MobileNet, TinyML, reduced computation precision, low-cost devices, and memory savings.
[52]	Wind farm control strategy using modifier adaptation for yaw control to improve power output.	Yaw control, modifier adaptation, wind farm management, power output optimization, and active yaw control.
[53]	Use of Support Vector Regression (SVR) to predict fatigue loads and power output under yaw control.	Support Vector Regression (SVR), fatigue load prediction, yaw misalignment correction, and accurate power output forecasting.
[54]	Impact of climate change and complex meteorological conditions on wind power prediction.	Climate change impact, meteorological variability, wind power prediction accuracy challenges, and extreme weather conditions.
[55]	Integration of grey models with machine learning to improve prediction accuracy in wind power forecasting.	Grey models, machine learning integration, nonlinear relationships, uncertainty handling, improved prediction robustness.
[56]	Machine learning models like LSTM and CNNs are used to analyze meteorological data and improve wind power forecasting.	LSTM (Long Short-Term Memory), CNN (Convolutional Neural Networks), time-series forecasting, and spatial and temporal data.
[57]	Machine learning algorithms for icing detection and prediction on turbine blades to improve operational efficiency.	Icing detection, predictive maintenance, sensor data analysis, turbine blade optimization, and operational efficiency.
[58]	Advancements addressing complexities in wind power prediction, supporting sustainable energy amidst climate change.	Sustainable energy solutions, climate adaptation, prediction model advancements, and addressing data complexities.
[59]	Use of XGBoost regression for short-term wind power prediction using SCADA system data.	XGBoost, regression modeling, SCADA system data, short-term forecasting, high-frequency data processing.
[60]	Impact of turbulence intensity on ultra-short-term wind power prediction; turbulence doesn't significantly improve accuracy.	Turbulence intensity, ultra-short-term wind power prediction, meteorological variables, model accuracy analysis.
[61]	Optimization methods like Particle Swarm Optimization (PSO) and the Fruit Fly Algorithm are used for improving wind power prediction models.	PSO (Particle Swarm Optimization), Fruit Fly Algorithm, hyperparameter tuning, optimization of machine learning models.
[62]	PSO is applied to optimize hyperparameters for machine learning models like neural networks and support vector machines.	PSO for hyperparameter optimization, neural networks, support vector machines, and tuning model parameters.
[63]	Use of the Fruit Fly Algorithm for optimizing parameters in wind power prediction models.	Fruit Fly Algorithm, optimization, parameter tuning, and improvement of prediction accuracy.

Table 9 (continued)

Citation	Existing Research	Main Features/Techniques
[64]	Benefits of PSO and the Fruit Fly Algorithm in improving model accuracy for wind power prediction.	Optimization efficiency, reduced prediction errors, model accuracy improvement, PSO, Fruit Fly Algorithm.
[65]	Ensemble learning methods like XGBoost and Random Forest to improve wind power prediction accuracy.	Ensemble learning, Random Forest, XGBoost, aggregation of model predictions, improved generalization, and accuracy.
[66]	Hybrid CNN-LSTM models to capture both spatial and temporal dependencies in wind power data.	Hybrid models, CNN-LSTM, spatial dependencies, temporal dependencies, complex data handling, and improved prediction accuracy.
[67]	Effectiveness of ensemble and hybrid models in managing uncertainty and variability in wind power data.	Ensemble learning, hybrid models, uncertainty management, data variability, robust predictions, and long-term forecasting.
[68]	Integration of machine learning and statistical approaches for interval forecasting in wind power prediction, LSTM enhances forecast accuracy.	LSTM (Long Short-Term Memory), statistical approaches, interval forecasting, uncertainty quantification, and forecast accuracy.

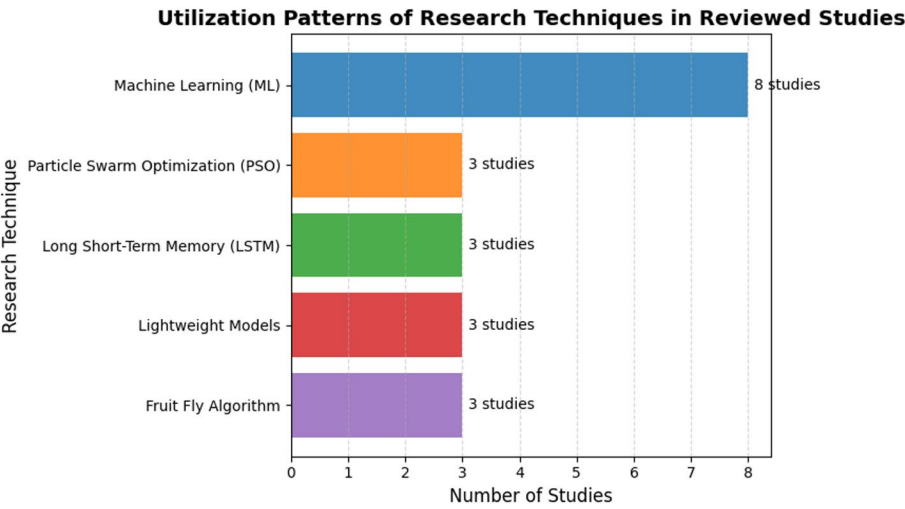


Fig. 6 Techniques employed in wind power prediction research frequently

(3 studies), PSO (3 studies), Fruit Fly Algorithm (3 studies), and lightweight models (3 studies) are the primary research approaches employed. In order to improve prediction accuracy, real-time performance, and computing efficiency, this study explores ensemble learning in conjunction with hybrid models, feature selection, and quantization techniques. The study demonstrates how researchers use deep learning models (LSTM, CNN) and optimization techniques (PSO, Fruit Fly Algorithm) in conjunction with real-time edge computing solutions for wind power forecasting (Fig. 6).

Figure 7 displays a line graph that illustrates the patterns in the application of various methods in wind power prediction studies over time. It shows how research has changed from 2015 to 2025 using approaches like machine learning, optimization (PSO, Fruit Fly), deep learning (LSTM, CNN), edge computing, and ensemble learning.

6 Key observations

Although advanced machine learning models like LSTM and CNN, along with ensemble techniques such as Random Forest and XGBoost, show strong potential for wind power forecasting, significant challenges remain in achieving real-time

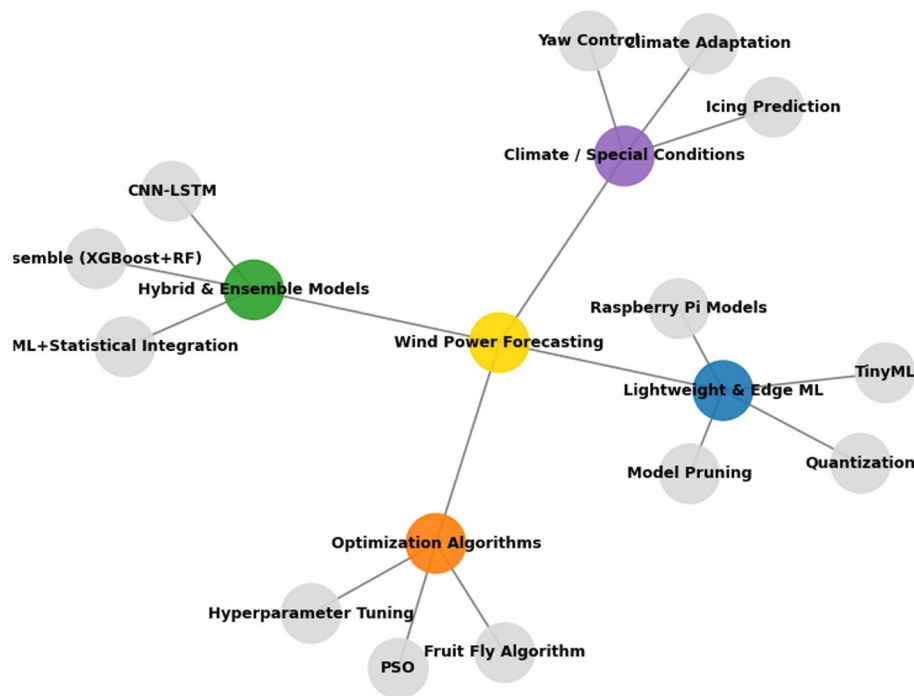


Fig. 7 Trends in Wind Power Forecasting Methods from 2015 to 2025

deployment—particularly on low-cost devices in remote locations. Further research is required to address operational issues caused by power fluctuations and system adjustments during extreme weather events. Equally crucial is analyzing how diverse climatic conditions influence model performance to ensure reliable predictions. Additionally, integrating wind power forecasting with other renewable energy sources could enhance global energy management. Techniques such as pruning, quantization, and effective feature engineering can help improve model efficiency and generalization, making them suitable for edge computing applications in remote wind farms.

6.1 Model applicability in different scenarios

While metrics like MAE, RMSE, and MAPE are useful for benchmarking, selecting the right wind power forecasting model also depends on the operational context. Key considerations include data availability, forecast horizon, geographic setting, computational capacity, and interpretability requirements. In data-rich environments such as large SCADA archives or offshore wind farms, deep learning models (e.g., LSTM, GRU, CNN, and hybrid CNN–LSTM) excel by exploiting high-resolution, multivariate inputs. In data-limited scenarios—such as sites with fewer sensors or newly commissioned farms—tree-based methods (Random Forest, XGBoost) and SVMs perform better due to their lower data requirements and strong generalization from engineered features. For ultra-short-term forecasts (<30 min), lightweight ML models or shallow neural networks deliver fast, accurate results with minimal latency, making them ideal for real-time control. LSTM and GRU models are more effective for short- to mid-term forecasts (hours to days) by capturing temporal dependencies in NWP-driven datasets. Long-term forecasting (months to years) benefits from hybrid approaches that combine physical and statistical models with ML to address seasonal and climatic variability.

Table 10 Appropriateness of deep learning and machine learning models for various wind power prediction situations

Scenario	Most Suitable Models	Rationale
Ultra-short-term, real-time	XGBoost, RF, shallow ANN	Fast inference, lower latency
Short-term (hour-ahead)	LSTM, GRU	Strong temporal modeling
Long-term, seasonal	Hybrid ML–NWP, CNN–LSTM	Capture climatic/seasonal signals
Data-rich	Deep learning models	Leverage high-volume, high-dimensional data
Data-scarce	RF, XGBoost, SVM	Perform well with fewer data, engineered features
Complex terrain	Gradient boosting, hybrid statistical–ML	Manage variability, nonlinearity
Low-resource deployment	RF, pruned/quantized DL	Resource-efficient operation

Geographic and climatic factors also influence suitability: CNN–LSTM and GRU models leverage the spatial-temporal stability of coastal/offshore sites, while gradient boosting and hybrid statistical–ML approaches better manage the irregularity of complex terrains. For locations affected by climate change, adaptive ML and transfer learning help minimize retraining as wind patterns evolve.

Computational resources guide deployment choices: deep learning fits well in high-performance cloud or HPC environments, whereas low-cost edge devices (e.g., Raspberry Pi) favor Random Forest, XGBoost, or optimized TinyML models through pruning and quantization. Finally, where interpretability is critical for operator trust, decision trees or linear models are preferable, while deep architectures require explainable AI techniques to make their “black box” predictions more transparent. Table 10 outlines the comparative suitability of these ML/DL approaches across forecast horizons, data scenarios, site conditions, computational limits, and interpretability needs.

7 Future research directions

Potential research directions can be ranked based on their viability, level of technological maturity, and anticipated time to implementation to direct future work in wind power forecasting.

7.1 Short-term priorities (1–3 years)

These research directions can be pursued using currently available methods, datasets, and computational resources. They include deploying advanced SCADA systems, nacelle lidar, and low-cost IoT sensors for real-time turbine and environmental data acquisition; enhancing data preprocessing pipelines for outlier detection, fault diagnosis, and missing data imputation to ensure high-quality inputs; applying hybrid feature selection techniques (e.g., Lasso with tree-based ranking) to optimise model accuracy and efficiency; developing lightweight ML/DL models through pruning, quantisation, and TinyML for edge computing in remote wind farms; and conducting robustness testing of models such as XGBoost and CNN–LSTM under high-frequency and incomplete data scenarios to simulate operational challenges.

7.1.1 Knowledge gaps and research priorities for high-penetration renewable grids

The rapid global shift toward high-penetration renewable energy systems introduces a unique set of challenges for wind power forecasting. While many advances have been

made in data-driven and hybrid modelling, several critical knowledge gaps remain that must be addressed to ensure reliable integration of wind energy into future power grids:

7.1.1.1 Forecasting under extreme variability and system stress Gap: Limited understanding of model robustness under sudden weather transitions, extreme events, and rapid ramping scenarios in high-renewable grids.

Direction: Research into resilient forecasting frameworks that leverage ensemble meteorological models, uncertainty quantification, and adaptive algorithms capable of real-time recalibration.

7.1.1.2 Multi-energy coupling and sector coupling forecasts Gap: Inadequate models for integrated forecasting across coupled renewable sources (wind, solar, hydro) when penetration exceeds 60–70% of supply.

Direction: Develop multi-vector energy forecasting systems that incorporate cross-resource correlations, storage states, and demand-side flexibility.

7.1.1.3 Sub-second to ultra-short-term grid support forecasts Gap: Insufficient attention to ultra-short-term (< 1 min) high-frequency forecasts needed to support inverter-based resources, grid-forming converters, and synthetic inertia provision.

Direction: Investigate high-frequency hybrid ML–signal processing approaches using phasor measurement units (PMUs), edge-AI forecasting, and wide-area measurement data.

7.1.1.4 Geographically distributed and edge-based forecasting Gap: Sparse research on coordinating forecasts from geographically dispersed assets in real time, especially in microgrids and distributed energy resource (DER) environments.

Direction: Develop federated and decentralized learning frameworks that integrate site-specific forecasts without centralized data sharing, ensuring cybersecurity and data privacy.

7.1.1.5 Explainability and trust in operational forecasts Gap: Lack of interpretable forecasting tools to support operator confidence in automated dispatch decisions in renewable-rich grids.

Direction: Advanced explainable AI for forecasting with visual analytics, attribution methods, and uncertainty bounds tailored to transmission operators and market participants.

7.1.1.6 Climate adaptation in forecasting models Gap: Forecasting methodologies often assume stationarity, with limited accounting for the impacts of long-term climate change on wind regimes.

Direction: Integrate climate projections and regime-shift detection into operational forecasting pipelines.

7.2 Medium-term priorities (3–5 years)

Medium-term priorities (3–5 years) involve advancing algorithms and integrating diverse modelling strategies, including transformer-based architectures with

temporal and spatio-temporal attention for long-horizon, multi-scale forecasting; physics-informed neural networks (PINNs) and hybrid CFD–ML models that fuse numerical weather predictions and fluid dynamics with ML/DL for better generalisation in complex terrains; multi-energy hybrid forecasting frameworks for jointly modelling wind, solar, storage, and demand to optimise grid operations; domain adaptation and transfer learning to quickly customise pre-trained models for new sites or extreme weather with minimal local data; and uncertainty quantification via probabilistic and ensemble methods to generate prediction intervals for market bidding and grid risk management.

7.3 Long-term priorities (5 + years)

Long-term priorities (5 + years) target transformational impacts through large-scale collaboration and advanced infrastructure, including fully integrated renewable energy forecasting ecosystems that unify AI-driven multi-energy generation, demand, storage, and market forecasting; federated and privacy-preserving collaborative learning to share model insights without exposing sensitive data; climate change-adaptive forecasting that incorporates climate projections to maintain accuracy amid long-term wind regime shifts; autonomous grid balancing systems that couple real-time forecasts with automatic corrective actions; and digital twins for wind farms that integrate live forecasting, asset health prediction, and maintenance optimisation.

By structuring research opportunities in this time-horizon framework, stakeholders can align investment and development roadmaps, ensuring that immediate operational gains are realised while paving the way for medium- and long-term innovation.

8 Conclusions

This review synthesises machine learning approaches for wind power forecasting, encompassing traditional ML, deep learning, ensemble, and hybrid models, along with data handling, optimisation, and evaluation metrics. Innovations include a category-based model classification, quantified impacts of data preprocessing, systematic algorithm comparisons, and coverage of special scenarios such as extreme weather and complex terrain. Limitations involve reliance on existing literature, absence of unified benchmarks, and limited discussion of hardware and economic factors. Future research should explore transformer-based models, multi-energy hybrid forecasting, spatio-temporal GNNs, transfer learning, and deployment-friendly lightweight architectures. By outlining both current advances and research needs, this work offers a roadmap for more accurate, robust, and operationally integrated wind power forecasting.

Abbreviations

ANN	Artificial neural network
ARIMA	Auto regressive integrated moving average
BO	Bayesian optimization
CNN	Convolutional neural network
CFD	Computational fluid dynamics
DTR	Decision tree regression
ESS	Energy storage system
FOA	Fruit fly algorithm
GBM	Gradient boosting machine
GRU	Gated recurrent unit
IoT	Internet of Things
Lasso	Least Absolute Shrinkage and Selection Operator
LR	Linear regression
LSTM	Long short-term memory
MAE	Mean absolute error

MAPE Mean absolute percentage error

Author contributions

Inam Ul Haq conducted the overall work, wrote the manuscript including drafting and implementation. Abhishek Kumar carried out the supervision and final review. Pramod Singh Rathore carried out the co-supervision and drafting the literature.

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Data availability

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Declarations

Ethics approval and and Consent to Participate

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